

COMP6721 Course Project: Decision Trees

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Abstract: This project explores the classification of museum images into two categories, indoor and outdoor, using both traditional machine learning models and deep learning techniques. A dataset of 10,000 images from the Places MIT dataset was preprocessed using feature extraction methods such as color histograms, edge detection, and grayscale intensities. Decision Tree, Random Forest, and Gradient Boosting models were trained on these features, with Gradient Boosting achieving 91.5% accuracy after hyperparameter tuning. The project also included a semi-supervised learning approach using pseudo-labeling, which yielded modest results (71.5% accuracy). Additionally, Convolutional Neural Networks (CNNs) were implemented to classify raw images directly. A no-pooling variant significantly outperformed all other models, reaching 97% accuracy. We present a full comparison of model performances, implementation strategies, and tuning processes, demonstrating how deep learning architectures can surpass classical models in visual classification tasks.

I. INTRODUCTION AND PROBLEM STATEMENT

Image classification is a fundamental task in computer vision that involves assigning labels to images based on their visual content [1]. It has wide-ranging applications in fields such as autonomous systems [2], surveillance, medical imaging [3], and cultural heritage preservation [4]. The ability to classify images accurately enables advancements in automated decision-making, object recognition, and large-scale data organization.

One of the primary challenges in image classification is handling variations in lighting, scene composition, and object presence, as differences in illumination and background clutter can impact object recognition [1]. Differences in illumination, texture, and background clutter can affect the ability of models to extract meaningful patterns. Robust preprocessing and feature extraction techniques are crucial to improving classification performance, particularly when dealing with diverse datasets.

The goal of this project is to classify museum images into two categories: *museum-indoor* and *museum-outdoor* using machine learning techniques, facilitating automated sorting and digital archiving of museum imagery [4]. This classification will facilitate automated sorting and analysis of museum imagery, which can be useful for applications in digital archiving, virtual museum experiences, and scene recognition systems.

The classification problem will be approached using both traditional and deep learning models. Initially, three supervised machine learning models: *Decision Tree*, *Random*

Forest, and *Gradient Boosting*, were selected for their ability to handle structured feature vectors derived from image data. *Decision Trees* offer an intuitive and interpretable baseline model [5], while *Random Forest* mitigates overfitting by aggregating multiple trees for improved generalization [6]. *Gradient Boosting* refines performance by focusing on misclassified instances through iterative updates [7].

A semi-supervised learning approach will also be explored, where *Decision Trees* are iteratively trained with labeled data, and high-confidence predictions on unlabeled data are incorporated for further refinement [8]. This approach aims to leverage both labeled and unlabeled data to enhance model performance, particularly when labeled samples are limited.

a) Related works

Image classification is a well-established problem in computer vision, with numerous techniques evolving over the past decades. Early approaches relied on handcrafted features such as Scale-Invariant Feature Transform (SIFT) and Histogram of Oriented Gradients (HOG), followed by classical classifiers like Support Vector Machines and Decision Trees [9]. Ensemble learning methods such as Random Forest and Gradient Boosting gained popularity due to their ability to reduce overfitting and improve predictive accuracy in high-dimensional feature spaces [10].

More recently, the rise of deep learning, particularly Convolutional Neural Networks (CNNs), has led to significant breakthroughs in image classification tasks. CNN-based architectures such as AlexNet [11], VGGNet [12], and ResNet [13] have demonstrated superior performance on large-scale image datasets like ImageNet by learning hierarchical feature representations directly from pixel data.

II. METHODOLOGY

B. Dataset Overview

The dataset used for this project is the Places MIT [14] dataset, specifically focusing on the *museum-indoor* and *museum-outdoor* categories. This dataset consists of 10,000 images evenly distributed across both classes. The images vary in lighting conditions and perspectives, requiring effective preprocessing to enhance model performance. [Figure 1]

C. Exploratory Data Analysis (EDA)

EDA was performed to understand the dataset composition and address any inconsistencies

1) Class Distribution

One of the first aspects analyzed was the class distribution, confirming that the dataset is balanced with 5,000 images per class. This ensures that the model does not favor one category over the other, reducing potential bias in classification.



Figure 1: Dataset overview

2) Resolution Consistency

All images in the dataset have a fixed resolution of 256x256 pixels, ensuring uniform input size for preprocessing and model training.

3) Color vs. Grayscale

This analysis revealed that color plays a significant role in differentiating indoor and outdoor museum settings. A color histogram analysis showed that indoor museums tend to have warmer tones (reds, yellows, and browns), while outdoor museums exhibit more blue and green hues due to the presence of natural elements such as sky and vegetation. [Figure 2]

Color Histogram: Indoor vs Outdoor Museums

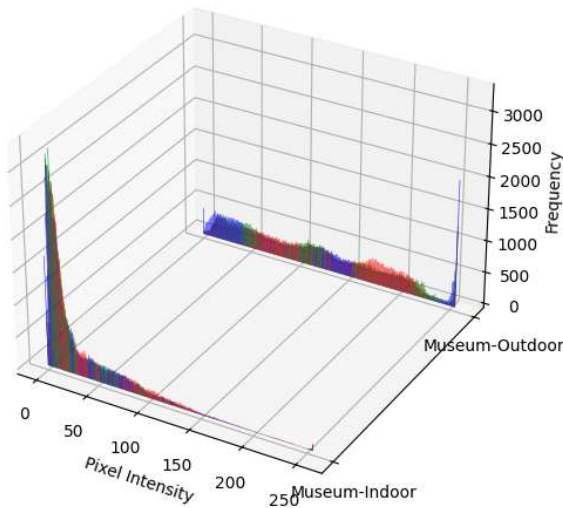


Figure 2: Color Histogram.

4) Edge Density

This analysis was conducted to explore differences in structural complexity between the two classes. It was observed that outdoor museum images generally contain more edges, reflecting the presence of buildings, trees, and sky. In contrast, indoor museum images tend to have fewer edges due to smooth walls and controlled lighting. This structural difference suggests that edge detection techniques could be valuable in feature extraction. [Figure 3]

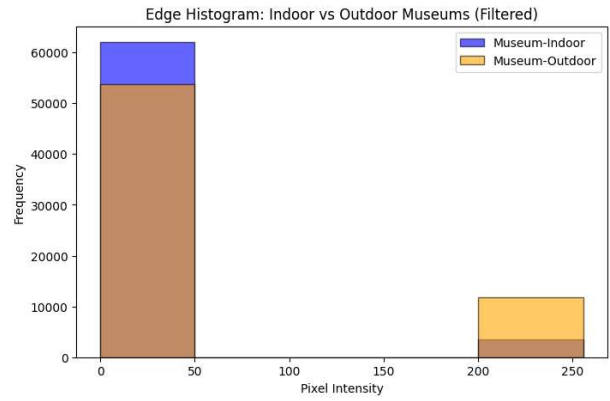


Figure 3: Edge Histogram

5) Brightness & Contrast Analysis

Indoor museums often have lower brightness due to artificial lighting, whereas outdoor museums are naturally illuminated by sunlight, leading to higher brightness levels. A histogram of brightness values confirmed this trend, with indoor images clustering in the lower brightness range and outdoor images showing a broader distribution. This suggests that brightness could serve as a distinguishing feature for classification. [Figure 4]

D. Image Preprocessing Steps

To ensure compatibility with machine learning models, multiple preprocessing techniques were applied:

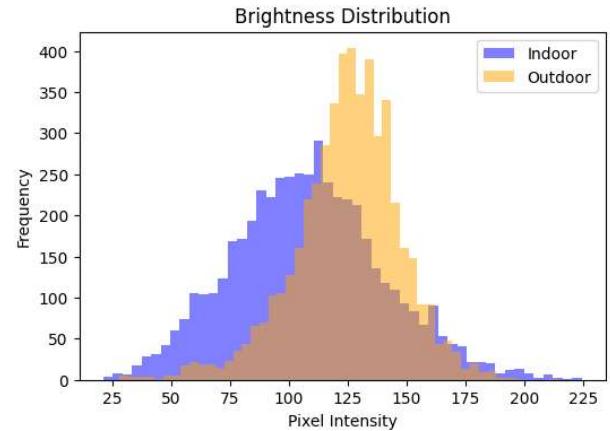


Figure 4: Brightness Distribution

1) Data Extraction and Filtering

- The dataset was provided in ZIP format, requiring extraction before processing.
- A text file (*train.txt*) contained file paths and corresponding class labels.
- Filtering was applied to select only museum-indoor and museum-outdoor images.

2) Standardizing Image Input

- All images were resized to 32x32 pixels due to memory constraints, ensuring efficient processing without exceeding computational limits.
- Image paths were checked for validity, ensuring no missing or corrupted files.

3) Feature Extraction for Model Compatibility

Since Decision Trees and Random Forest models require structured numerical data, images were converted into feature vectors using three complementary methods:

- *Flattened Pixel Values*: Each image was converted to grayscale and flattened into a 1D vector of pixel intensities.
- *Color Histograms*: Extracted from the HSV color space using 8 bins per channel, providing a 512-dimensional feature vector.
- *Edge Detection (Canny Operator)*: Captured structural patterns by detecting edges, which were then flattened into a 1D feature vector.

4) Caching Preprocessed Data

Since feature extraction is computationally expensive, processed feature vectors were cached using *pickle* to prevent redundant calculations and speed up training.

E. Model Selection and Justification

For this project, three supervised machine learning models were selected: Decision Tree, Random Forest, and Gradient Boosting; each offering unique strengths in handling structured feature vectors.

1) Decision Tree Classifier

Decision Trees were chosen as a baseline model due to their simplicity and interpretability. They provide a clear visualization of decision boundaries by recursively splitting features. However, they tend to overfit complex datasets, limiting their generalization ability. [5]

2) Random Forest Classifier

Random Forest was selected to mitigate the overfitting issue of Decision Trees. By averaging multiple decision trees trained on random subsets of data, Random Forest provides better generalization and is less sensitive to noise. Additionally, it allows for feature importance analysis, which helps in understanding which extracted features contribute most to classification accuracy. [6]

3) Gradient Boosting Classifier

Gradient Boosting was included as it iteratively improves performance by focusing on misclassified samples. Unlike Random Forest, which averages multiple weak models, Gradient Boosting sequentially refines its predictions, often resulting in better accuracy. However, it requires careful hyperparameter tuning and is computationally expensive. [7]

4) Convolutional Neural Networks (CNN)

Convolutional Neural Networks (CNNs) were employed to classify museum images using raw image inputs instead of extracted features. The baseline CNN architecture consisted of three convolutional layers with *ReLU* activation, followed by max-pooling and fully connected layers. Variants included a model without pooling layers and another with an additional convolutional layer to deepen the network. CNNs are well-suited for image classification tasks due to their ability to automatically extract hierarchical spatial features and detect patterns such as textures and edges.

All CNN models were trained using the *Adam* optimizer, known for its adaptive learning rate and efficient convergence properties, especially in high-dimensional parameter spaces. The models were optimized using *CrossEntropyLoss*, appropriate for multi-class classification problems.

III. RESULTS

F. Model Training and Evaluation

All experiments were conducted using Python notebooks in Google Colab. *Decision Tree*, *Random Forest*, and *Gradient Boosting* were implemented using *scikit-learn* and trained on the CPU. They were trained using the extracted feature vectors (color histograms, grayscale intensities, and edge detection data).

For deep learning experiments, including the various *CNN* architectures, the environment was switched to *CUDA-enabled GPU* to significantly reduce training time. The models were implemented using *PyTorch*, and trained on the full dataset after appropriate preprocessing and transformation.

Model validation was performed using a hold-out test set for both classical and CNN models to ensure consistent evaluation.

DECISION TREE, RANDOM FOREST, AND GRADIENT BOOSTING

1) Baseline Model Performance

The initial models were trained using default hyperparameters, and their performance was evaluated using accuracy, precision, recall, and F1-score. The table below presents the results:

Model	Accuracy	Precision	Recall	F1	Training
DT	0.775	0.775	0.775	0.775	2 min
RF	0.875	0.875	0.875	0.875	1.5 min
GB	0.865	0.865	0.865	0.865	6 min

Table 1: Baseline model performance

As expected, Random Forest and Gradient Boosting outperformed the Decision Tree model, as they benefit from ensemble learning, which mitigates overfitting and improves generalization. Gradient Boosting further enhanced performance by iteratively correcting misclassified samples.

2) Hyperparameter Tuning

To further optimize performance, *RandomizedSearchCV* was used for hyperparameter tuning instead of *GridSearchCV*. Initially, *GridSearchCV* was considered, but it performed an exhaustive search over all parameter combinations, making it computationally expensive and time-consuming. *RandomizedSearchCV* was chosen as a more efficient alternative, allowing us to explore a wide range of hyperparameters while significantly reducing training time.

By using *RandomizedSearchCV*, we achieved a balance between optimization efficiency and computational cost, ensuring that training time remained manageable while still finding near-optimal hyperparameter values.

The best hyperparameter configurations for each model were identified and used to retrain the models, leading to the following improved results:

- **Decision Tree:** *max_depth* [10], *min_samples_split* [5], *min_samples_leaf* [1]
- **Random Forest:** *n_estimators* [200], *max_depth* [None], *min_samples_split* [10]

- **Gradient Boosting:** $n_estimators$ [200], $learning_rate$ [0.2], max_depth [10]

Model	Accuracy	Precision	Recall	F1	Search
DT	0.81	0.81	0.81	0.81	6 min
RF	0.87	0.87	0.87	0.87	6 min
GB	0.915	0.915	0.915	0.915	5.5 hours

Table 2: Tuned model best performance

The tuning process significantly improved classification accuracy across all models, demonstrating the importance of optimizing hyperparameters for better generalization.

3) Semi-Supervised Learning Implementation

Since labeled data is often scarce, a semi-supervised learning approach was implemented using iterative pseudo-labeling. The Decision Tree model was initially trained on a small subset (20%) of labeled data. Predictions were then made on the unlabeled 80%, and high-confidence predictions (probability $\geq 85\%$) were added to the labeled dataset for further refinement. This iterative process was supposed to be repeated for up to 10 iterations if the precision was out of the specified range. However, in our case, the process concluded after the first iteration, as the achieved precision was already within the specified range, eliminating the need for further iterations.

Model	Accuracy	Precision	Recall	F1	Training
DT	0.715	0.715	0.715	0.715	1 min

Table 3: Semi-Supervised model performance

The semi-supervised approach successfully leveraged unlabeled data to improve performance, but the gains were limited compared to the fully supervised models.

CONVOLUTIONAL NEURAL NETWORKS CNN

4) Baseline model performance

To evaluate the performance of deep learning approaches on the same classification task, several Convolutional Neural Network (CNN) architectures were trained using the original image dataset. The models were compared based on structural modifications, including the removal of pooling layers and the addition of an extra convolutional layer.

Model	Accuracy	Precision	Recall	F1	Training
CNN: Baseline	0.91	0.91	0.91	0.91	<1 minute
No pooling	0.97	0.97	0.97	0.97	<1 minute
Extra conv layer	0.91	0.91	0.91	0.91	<1 minute

Table 4: CNN model performance

The no-pooling architecture outperformed the others, achieving the highest classification metrics across the board. This suggests that retaining spatial resolution throughout the convolutional stack can be beneficial for fine-grained indoor/outdoor scene recognition. The addition of a convolutional layer did not lead to performance gains, indicating that deeper is not always better for this task.

5) NN Hyperparameter Tuning

To further improve performance, a grid search was conducted over a range of learning rates and batch sizes with the following values:

- **Learning Rates:** [1e-1, 1e-2, 1e-3, 1e-4]
- **Batch Sizes:** [16, 32, 64]

These values were selected based on commonly recommended ranges in deep learning practices. Lower learning rates allow for more stable convergence. Higher learning rates were included to observe the model's sensitivity to aggressive weight updates. The batch sizes were chosen to balance between training stability and memory efficiency; 16 being more accurate with slower updates, while 64 offers faster training but with less gradient variability.

The best configuration for the baseline CNN model was identified as a *learning rate* of **0.001** and a *batch size* of **16**, achieving the best accuracy with a good trade-off between training speed and generalization.

Model	Accuracy	Precision	Recall	F1	Training
CNN Base tuned	0.9	0.9	0.9	0.9	1.77 s

Table 5: hypertuned CNN model performance

The tuning process validated the baseline architecture's efficiency, offering slight gains in training stability and convergence while maintaining strong performance. Despite the tuned model showing marginally lower metrics than the no-pooling variant, it achieved competitive results with very fast training time.

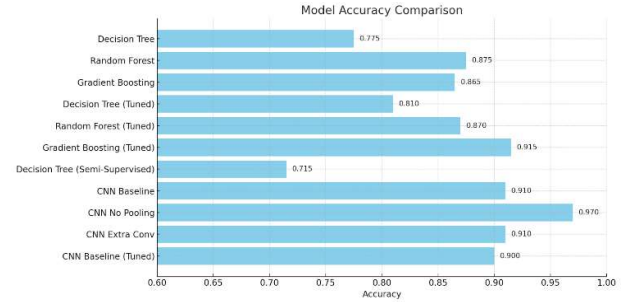


Figure 6: Model Accuracy Comparison

IV. CONCLUSION

Training our data on the baseline models results in the best-performing model of Random Forest having **87%** accuracy and recall when compared with other models. By changing the hyperparameters, the best-performing algorithm turned out to be Gradient Boosting with accuracy and recall of up to **91.5%**. Training our data on the baseline models results in the best-performing model of Random Forest having **87%** accuracy and recall when compared with other models.

By changing the hyperparameters, the best-performing algorithm turned out to be Gradient Boosting with accuracy and recall of up to **91.5%**.

The introduction of CNN-based models demonstrated the potential of deep learning in this classification task. The no-pooling variant of the CNN significantly outperformed all traditional models, achieving a peak accuracy and recall of **97%**, while the baseline and extra-conv CNN variants both maintained strong performance at **91%**. Hyperparameter tuning of the baseline CNN led to stable results: **90%**, confirming that even with minimal training time, CNNs can match ensemble models in effectiveness.

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